Fiscal Year:	FY 2005	**Task Last Updated:**	FY 04/07/2005
PI Name:	Angelaki, Dora Ph.D.		
Project Title:	Multisensory Interactions to Discriminate Gravity From Translational Acceleration		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Organismal biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology (2) Neurobiology		
Space Biology Special Category:	(1) Cell Culture (2) Translational (Countermeasure) Potential		
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Zip Code:	63110-1093	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Biology 03-OBPR-03
Start Date:	07/01/2004	**End Date:**	06/30/2007
No. of Post Docs:	3	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Dickman, J David (Washington University Medical School)		
Grant/Contract No.:	NNA04CC77G		
Performance Goal No.:			
Performance Goal Text:			
	Throughout the history of the manned space flight program, microgravity has produced vestibular-related symptoms that result in personal discomfort and a loss in crew performance. The symptoms subside within several days of microgravity exposure, suggesting that interactive multisensory mechanisms may be responsible for the initiation of symptoms and the subsequent adaptation. Specifically, the processing and detection of gravity and translational accelerations is one of the most fundamental issues for understanding vestibular sensorimotor control in normal gravity, as well as in altered gravity environments. In order to better understand the nature of multisensory signals in detecting gravity, we will characterize how gravity and translational accelerations are encoded by vestibular neurons in the cerebellum of alert primates. Unless these multi-sensory interactions are better understood in normal gravity, our understanding of neurovestibular adaptation to altered gravity and the development of relevant countermeasures remains largely		

Task Description:	incomplete. The proposed studies aim at filling an important gap in existing knowledge through state-of-the art electrophysiological and computational techniques. Aim I: Investigate the vestibular convergence patterns in the rFN and NVU: We propose to use combinations of three-dimensional (3D) rotations and translations in order to characterize the otolith/semicircular canal convergence patterns in the rFN and NVU. Specifically we propose: (1) To examine if there are populations of rFN and NVU neurons with properties similar to those found in the VN (Canal-Only, Otolith+Canal and Otolith-Only cells). (2) To quantify differences and/or similarities in the 3D distributions of rotational maximum sensitivity vectors using earth-vertical axis (EVA) rotation (canal only stimulation), as well as in the spatial distributions of translational sensitivity vectors (otolith only stimulation) between convergent and non-convergent cells. (3) To compare EVA rotation and translational responses with cell firing rate modulation during earth-horizontal axis (EHA) rotations in order to examine the pattern of otolith/canal convergence in the rFN and NVU. Our working hypothesis is that the rFN and NVU circuitry is specific to the processing of linear acceleration. Thus, unlike the VN where a quarter of the sampled neuronal population was identified as Canal-Only cells, we hypothesize that we will encounter very few, if any, Canal-Only neurons in the rFN and NVU. Aim II: Test if rFN and NVU neurons discriminate dynamic tilts relative to gravity from translational components of linear acceleration: Although primary otolith afferents respond equally to translational acceleration and dynamic tilts relative to gravity, central vestibular neurons could selectively encode the translational component of the motion. We will quantitatively test this hypothesis using protocols with combinations of translational and tilt motion stimuli. We hypothesize that a subpopulation of rFN and NVU neurons, including all neurons antidromically identified as projecting to the VN, will selectively respond to translational head motion and ignore changes in head orientation relative to gravity. Aim III: Investigate the coordinate system (head vs. body-fixed) for encoding translational motion by rFN and NVU neurons: Primary otolith afferents encode linear acceleration in a head-fixed coordinate system. However, spatial orientation, postural control and self-motion awareness might require that there exist central neurons encoding translational motion of the body in space. In these studies we will investigate the influence of static neck inputs in the processing the otolith information by systematically varying both the direction of sled motion relative to the body and the static orientation of the head relative to the trunk. Because both coordinate systems might be important for downstream processing, we hypothesize that translation-coding rFN and NVU neurons that project to the VN include both head-, as well as body-fixed coordinate systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	We have made significant progress in all three aims of the grant. In particular, we have characterized the otolith/canal convergent properties in the rFN (aim 1). We have also used combinations of tilts and translation to test if central vestibular neurons discriminate tilt from translation (aim 2). Finally, we have compared the coordinate frames of coding motion in the rFN and VN. The accomplished work has resulted in four published papers (2 in the Journal of Neurophysiology, one in the Journal of Neuroscience and the fourth in Nature). Two additional manuscripts are currently under review.
Bibliography Type:	Description: (Last Updated: 04/21/2017)
Abstracts for Journals and Proceedings	Newlands, S.D., Shaikh A. G., Green A.M., Dickman, J.D. and Angelaki, D.E. "Semicircular canal inactivation abolishes tilt-translation discrimination property of fastigial and vestibular nuclei neurons" Soc. Neurosci. Abstr. 30, San Diego CA. , Nov-2004
Abstracts for Journals and Proceedings	Shaikh A. G., Ghasia F.F., Dickman, J.D. and Angelaki, D.E. "Properties of cerebellar fastigial neurons during translation, rotation and eye movements" Soc. Neurosci. San Diego CA. Soc. Neurosci. Abstr. 30 , Nov-2004
Abstracts for Journals and Proceedings	Shaikh A. G., Newlands, S.D., Dickman, J.D. and Angelaki, D.E. "Semicircular canal inactivation abolishes tilt-translation discrimination property of fastigial and vestibular nuclei neurons" Association for Research in Otolaryngology Abstr., New Orleans LA. ARO 2005 , Feb-2005
Articles in Peer-reviewed Journals	Green A.M. and Angelaki D.E. "An integrative neural network for detecting head orientation and movement" J. Neurophysiol. 92(2):905-925 , Aug-2004
Articles in Peer-reviewed Journals	Shaikh A.G., Meng H. and Angelaki, D.E. "Multiple reference frames for motion in the primate cerebellum" J. Neuroscience 24 4491-4497 , May-2004
Articles in Peer-reviewed Journals	Angelaki D.E., Shaikh A.G., Green A.M. and Dickman, J.D. "Neurons compute internal models of the physical laws of motion" Nature 29;430(6999):560-564 , Jul-2004
Articles in Peer-reviewed Journals	Shaikh A.G., Ghasia F., Dickman J.D. and Angelaki, D.E. "Properties of cerebellar fastigial neurons during translation, rotation and eye movements" J. Neurophysiol. 93(2):853-863. , Jan-2005